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REVIEW ARTICLE

Organ-On-A-Chip Technology: An In-depth Review of Recent Advancements and Future of Whole Body-on-chip

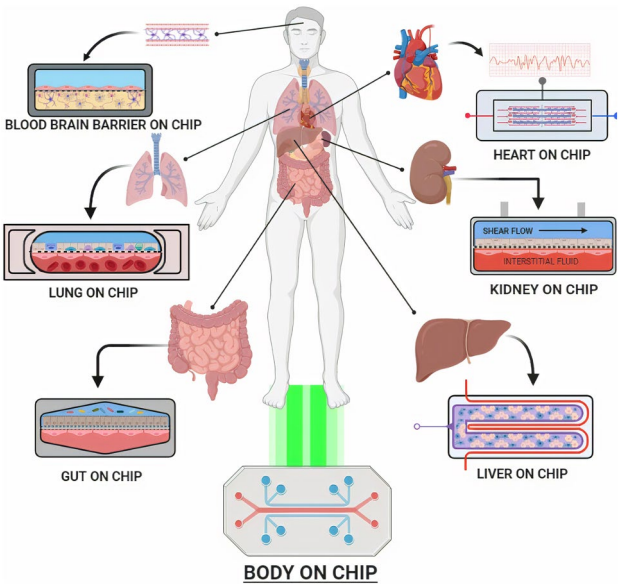
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Abstract

One of the leading challenges facing the pharmaceutical industry today is the 'high attrition rate problem', with a large number of drug candidates being abandoned before reaching the clinical stage. The high failure rate is not only costly for the pharmaceutical industry but also hinders the access of patients to better treatment options. This ever-widening gap can be attributed to the current in-vitro and animal testing techniques for predicting human responses to drug candidates. Two-dimensional (2D) in-vitro cell culture fails to recapitulate key physical and biochemical cues while phylogenetic differences between animal models and humans in key physiological systems make data generated by these platforms unreliable. There have been several solutions proposed but one that has been gaining a lot of prominence are microfluidic-based microsystems able to mimic organs, called 'Organ-On-A-Chip' devices and is built upon the principles of microfluidics, material science, and cell biology. Here we delve into the ongoing research involving this platform demonstrating its key strengths and the application of this technology in several key stages of drug discovery such as target identification and high throughput screening. We also discuss the potential, future, and key obstacles that lie in the universal adoption of this technology.

Graphical Abstract



Keywords Organ-On-A-Chip · Microfluidic chip · Microfluidics · Physiological model · 3D culture · Toxicity profile

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1 Introduction

According to a report from the European Commission (EC) to the Council and The European Parliament in 2019, between 2015 and 2017, 9.59 million (2015), 9.82 million (2016), and 9.39 million (2017) animals were used for research and testing. The number of animals used remained below 10 million, significantly lower than 11.5 million reported in 2011. This reduction in numbers can be attributed to the ban on animal testing and many alternative methods of testing reaching regulatory acceptance. However, the main areas of animal use remained unchanged, with most uses being in basic research (45%), followed by translational/applied research (23%) and regulatory use (23%). The development of a drug takes on average 13.5 years and $2.5 billion in funding; 92% of drugs fail in human clinical trials and never reach the market. Despite successful pre-clinical trials, 85% of early clinical trials for novel drugs fail; of those that survive through to phase III, only 50% become approved for clinical use [1]. The FDA's Adverse Event Recording System has received over 2.3 million reports of adverse drug reactions for 6000 registered compounds over 30 years, beginning from 1969. Over 75 drugs were removed from the market due to their unpredicted side effects in this period. Additionally, 11 more were given special restrictions. Roughly out of every ten drugs entering clinical trials, only one passes for human treatment [2]. The main reason behind the high rate of failure is the biological difference between humans and animals. Crucial genetic, molecular, immunologic, and cellular differences between humans and mice make animal models inaccurate in determining drug candidates' efficacy [3]. Various studies have revealed that animal studies overestimate possibility of treatment by 30% and is a consequence of results being unpublished [4]. 4000 + genes in humans and mice were analysed, and researchers found that transcription factor binding sites differed between the species in 41% to 89% of cases [5]. Also, stress is a crucial factor in caged animals in a laboratory environment [6]. Most animals are bred in sterile conditions causing them not to develop an immune response making it impossible to model diseases [7]. The gut microbiome also affects drug metabolism, but the difference in this biome across species results in inaccurate results [8]. Animal testing also requires large amounts of funding to account for breeding, housing, and disposal burdens. Also, they carry the ethical burden of sacrificing animals in experiments and stress human researchers carrying out these experiments.

These factors have created the need for newer in-vitro testing platforms that could generate reliable data in terms of the efficacy and toxicity of new drug candidates. These platforms need to be sensitive as many new novel compounds have a narrow safety window. However, current in-vitro techniques involving the culturing of cells in petri dishes or multi-well plates are highly inadequate. Firstly, there is a reduction in cell sensitivity when cultured outside the body and can be attributed to the lack of several important physical and biochemical cues. Cells in-vivo are supplied with blood flow, which provides oxygen and nutrients while preventing the pooling of metabolites and waste material generated by cells. However, this is impossible to mimic in well plates and petri dishes, which often leads to exhaustion of nutrients and accumulation of toxic wastes, preventing the cells from functioning at full capacity and making long-term analysis impossible. Also, the detection of essential metabolites or chemical messengers produced by the cells becomes challenging as they become too diluted to allow detection. Drug absorption analysis in well plates is not accurate, as the absence of fluid flow often arrests diffusion, leading to lower values being generated. Fluid shear is often a crucial physical cue for histodifferentiation of cells, such as kidney tubule cells. Further, drugs not only interact with the target tissue but also with every other tissue in the body. The in-vivo environment is highly dynamic, with changes in one tissue system affecting other tissue systems, as in the case of metabolite induced cytotoxicity, making it crucial that different types of tissues be allowed to interact with one another to better predict drug outcome in the human body.

Organ-on-Chip technology, also referred to as Microfluidic Systems, offers to be a unique solution to the testing dilemma and has been generating interest worldwide.

Microfluidic systems can be defined as systems that can manipulate small (10^{-9} to 10^{-18} L) amounts of fluids. These systems generally consist of channels with dimensions lying in a range of tens to hundreds of a micrometre. Microfluidic systems were initially used in the analysis, presenting several advantages such as high sensitivity and resolution and shorter duration of the analysis. These advantages can be attributed to the small microchannels, allowing fluid to flow under laminar flow, giving the unique ability to control concentration in space and time. The quantities of samples and reagents required are minimal, reducing costs and the footprint for the analytical devices [9].

The Organ-On-Chip or Microfluidic systems consist of the following components, (1) The chip, designed for each unique application based on the fundamentals of microfluidics. These chips are generally fabricated from elastic polymers such as—poly(dimethylsiloxane) or PDMS, a transparent and soft elastomer. Each chip is designed for its unique application with chambers to hold tissues and microchannels connecting these chambers allowing cross-talk and fluid flow through the different types of tissues. They are generally fabricated using photolithography, and

soft lithography techniques; (2) living tissue, which generally consists of cells obtained from immortalized cell lines; (3) pumps and valves used to generate fluid shear and control flow through the device. Peristaltic pumps, syringe pumps, and pneumatic valves are used commonly. Furthermore, there has been a shift towards pumpless devices using gravity to generate fluid shear as they not only are they cheaper but also do not face the problem of introduction of air bubbles in the system generally faced with pumps. (4) Detection, which generally involves collecting fluid flowing out of the chip in reservoirs for further analysis. There have been advancements in separation and detection, allowing it to be carried out on-chip.

Microfluidic devices solve the majority of challenges facing current in-vitro systems. The application of a continuous fluid shear force ensures replenishment of nutrients and oxygen, with metabolites and other toxic waste materials being flushed away. This dramatically enhances the functioning of cells and ensures the survival of cells for long-term studies. Also, it helps induce histochemical differentiation of cells requiring fluid shear as a physical cue, such as epithelial cells which require pressure for the induction of chemical and gene regulatory responses. The pressure exerted on epithelial cells is crucial for the induction of chemical and gene regulatory responses of endothelial cells [10]. Determination of drug absorption would be more accurate on a microfluidic chip as flow rates can be adjusted to mimic in-vivo conditions, and the continuous removal of absorbed molecules would ensure a constant diffusion rate. A chip can be designed with multiple interconnected cellular compartments where a compound's effect can be studied on multiple systems simultaneously. This design also helps model the dynamic environment found in-vivo and help study metabolite induced cytotoxicity. As these systems can function with small volumes of a medium, it prevents the dilution of essential metabolites and biochemical messengers whose detection could shed new light on physiological reactions. This ability could also help model disease states and help understand a disease's progression in the human body (as shown in Fig. 1).

The advantages discussed above not only makes Organ-On-Chip an effective tool, but the platform also has potential to reduce R&D costs. Experts estimate a reduction of 10–26% in R&D costs, with highest applicability in the preclinical phase followed by the optimisation phase [11].

In the course of this review article, we would be going through several applications of this platform, demonstrating its flexibility to perform in different scenarios. The methodology, results, and critical observations would be discussed, allowing the reader to recognize the platform's strengths. It is crucial to note that these examples only represent a small fraction of all the research being done worldwide adopting this platform, enabling the reader to appreciate the true potential of this platform in being the testing standard of the future.

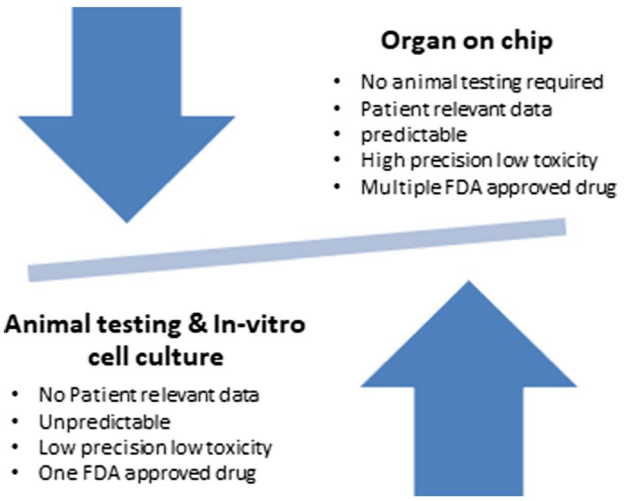


Fig. 1 Advantages of Organ on chip over *In-vitro* animals and cell lines study

2 Microfluidic Devices for Disease Modelling

Modelling of diseases is a crucial step to help identify possible targets for drug therapy. Disease states in vivo involve a complex cascade of reactions involving several different chemical messengers. The dynamic nature is impossible to be mimicked in current static in-vitro cultures. Hence animal models are generally used for this purpose, but the biological difference between humans and animals does not make the data generated entirely reliable. Also, it is not possible to continuously monitor, and the limited number of samples that can be drawn from an animal makes it difficult to understand the complete cascade. Some chemical messengers have a short life and would only be detected with continuous monitoring. Microfluidic devices, on the other hand, can better mimic the dynamic environment and enable researchers to continuously monitor the progress of the disease. The small working volumes ensure there is no dilution of critical chemicals and help gather new previously unknown information.

2.1 Microfluidics Devices for Modelling Neurological Disorders

2.1.1 Parkinson's Disease

Parkinson's disease is characterized by morphological changes in the transverse sections of the brainstem, with loss of the darkly pigmented area in the substantia nigra

pars compacta (SNpc) and locus coeruleus occurring in the majority of the cases. This loss in pigmentation is caused by the loss of dopaminergic (DA) neuromelanin-containing neurons in the SNpc and noradrenergic neurons in the locus coeruleus [12]. The development of in-vitro cellular models of the disease has been a significant challenge for research into this disorder.

Kane [13] developed a fully automated cell culturing platform using three-dimensional microfluidic devices capable of long-term maintenance and monitoring of cells to assess electrophysiology and cellular activity. Induced pluripotent stem cells (iPSCs) were used to generate human neuroepithelial cells (hNESCs), differentiated into dopaminergic neurons, and used for in-vitro modelling. PINK1-mutant human neuroepithelial stem cells were also cultured using this same procedure. Automation was achieved using a Pelican automated cell culture observatory, with Laboratory Automation Control Suite (LACS) used to create protocols to automate the different steps in the experiment.

Calcium imaging assay and immunofluorescence staining assay were used to analyse cultured cells. Calcium imaging assay involved the use of permeant Fluo-4 AM immunostaining was carried out using neuronal markers class 3 beta-tubulin (TUBβIII) and tyrosine hydroxylase (TH).

iPSC derived and PINK-1 mutant human neuroepithelial cells were successfully differentiated into midbrain-specific dopaminergic neurons, with the entire process being automated. The differentiated neurons can be maintained in the microfluidic devices for at least 100 days. It opens up several opportunities for researchers to gain new biochemical information. Patient-specific therapy is possible as it is possible to culture cell samples from the patient to gain new insight and plan the best possible treatment strategy.

2.1.2 Alzheimer's Disease

Alzheimer's disease is a common neurodegenerative disorder affecting 27 million people worldwide, characterized by memory loss and cognitive decline [14]. A prominent histopathological feature of this disorder is the formation of senile plaques made up of β-amyloid (Aβ), a peptide consisting of ~ 42 amino acid residues. It is produced by a membrane-bound amyloid precursor protein (APP) and proteolytic actions of β- and γ-secretases [15]. However, there exists a lack of knowledge of underlying mechanisms and analytical tools for screening drug candidates that can prevent aggregation of amyloid peptides.

Lee [16] developed a novel microfluidic screening platform that can not only allow for high throughput screening but also act as an analytical tool to analyse various environmental factors that can potentiate the formation of senile plaques. Microchannels were fabricated from poly(dimethylsiloxane) (PDMS) using photolithography. Next, the microchannels were seeded with Aβ42 seeds to form aggregates within the channels. Amyloid aggregation was bought about by perfusing Aβ42 feeding solution in the seeded microchannels at a constant rate of 5μL/h and incubating for 24 h at 37 °C, amorphous aggregates along with Aβ fibrils covering the entire inner surface was obtained.

This microchip was then used for the high throughput screening of several different small-molecule inhibitors that prevent amyloidogenic peptides' aggregation. Metal ions which have been known to play a role in the abnormal aggregation of Aβ peptides and the progression of Alzheimer's were also investigated [17-19]. Four metal ions Fe^{+3}, Cu^{+2}, Zn^{+2}, and Al^{+3} was tested. Fibrillation of Aβ was enhanced in the presence of Al^{+3} and Fe^{+3} ions, while Cu^{+2} and Zn^{+2} caused the formation of amorphous aggregated rather than fibrillar ones. These results conform with previous research.

Park [20] developed a microfluidic device consisting of arrays of concave microwells. Three dimensional (3D) neurospheroids were cultured in these microwells under constant perfusion using an osmotic pump to mimic the dynamic nature of the interstitial space *in-vivo*. The neurospheroids cultured in the chip showed a stronger and more well-connected neural network than those cultured under static conditions. The effects of Amyloid-β were also investigated using this platform.

These microfluidics-based systems can replicate Aβ aggregation quicker with more uniform fibril formation and could help improve the *in-vitro* modelling of neurodegenerative diseases. They can carry out high throughput screening, which can be used to identify other possible anti-amyloid agents and can also be used to investigate other possible factors that could play a role in formation of Aβ peptides.

2.1.3 Characterization of Neuropeptide Release

Neuropeptides are essential chemical messengers and are involved in communication between cells in the central and peripheral nervous system. They are synthesized within neurons and stored in vesicles and are released only in response to specific stimuli. They affect various physiological processes such as learning and memory, food intake, pain, and disease [21]. Neuropeptides also help establish and renew neuron-neuron interactions, and identifying these neuropeptides and conditions under which they are released can shed new light on formation and repair in neural networks. Current in-vitro techniques end up diluting released peptides, making their characterization difficult.

Croushore [22] developed a microfluidic device to support neuronal cultures and employed microvalves to stimulate these neuronal cultures selectively. The device was designed to have two layers, a flow channel and a pressure channel layer. The pressure channels were connected to an in-house built pneumatic solenoid valve system. Bag cell

neurons were used for this experiment, obtained from the abdominal ganglia of Aplysia californica.

Elevated levels of K^+ ions were used to stimulated the release of neuropeptides, with K^+ ions acting as a secretagogue, with the released neuropeptides detected by Matrix-assisted laser/desorption ionization (MALDI)—time-of-flight (TOF)-MS. Long term analysis of these bag cell neurons revealed that they could be maintained in the device for 12 days with regular media changes, allowing for multiple-day stimulation tests.

This experiment shows the ease of application of the microfluidic platform. The chip can support different types of neuronal cultures and can be coupled with different analytical tools, like the approaches taken by Zhong [23] and Rubakhin [24]. The difference in neuropeptide release due to varying exposure times revealed by this platform demonstrates its potential in expanding our knowledge about factors responsible for neuropeptide release. The sensitivity of the platform can also enable us to discover peptides previously undetermined and help us uncover molecules that promote neural growth and their potential use in reversing brain damage.

2.1.4 Amyotrophic Lateral Sclerosis (ALS)

Amyotrophic lateral sclerosis (ALS) is a fatal neurodegenerative disease affecting motor neurons. There is degeneration of cortical and spinal motor neurons, with its etiology being unknown [25]. Nonetheless, astrocytes have been implicated in disease pathogenesis, and hence it is crucial to understand the interactions between the neural cells and astrocytes. Current ALS studies are carried out using mouse models [26, 27]. However, they cannot be used to analyse drugs because of biological differences and ethical concerns [28]. Current in-vitro techniques involve the culturing of astrocytes as a feeder layer and neurons being plated over them [29]. These cell types are then exposed to various drugs [30]. However, this is an oversimplification of the in-vivo environment, and a platform that can mimic the complex in-vivo conditions accurately is required.

Kunze [31] developed a microfluidic device to support astrocyte-neuron cell culture. Genetically modified astrocytes were used that overexpressed either a human wild-type (WT) or a mutated form of the superoxide dismutase enzyme 1 (SOD1). The device was designed to prevent any direct cell to cell contact, allowing evaluation of neuronal cell's response to being in the vicinity of astrocytes. The microfluidic design consisted of three main parts, (1) Main channel with inlet channel, (2) Side perfusion channel, and (3) 24 junction channels connecting the main channel and perfusion channel. The astrocytes were placed in the perfusion channel, while the neural cells were placed in the main channel. The junction channels separated these two cell chambers by a distance of 0.94 mm. This distance maintained between the cells was essential to separate cell–cell interaction from metabolic interactions. The cells were placed close enough to allow the study of extracellular signals, the role of specific concentrations, and the influence of gradients of metabolites and prevent cell–cell coupling.

The cells were then exposed to oxidative stress by increasing extracellular glutamate concentrations levels above physiological levels. A 45% decrease in neuronal cell density was found when neurons were co-cultured with SOD-mutant astrocytes. This is because the SOD enzyme reduces reactive oxygen species produced by neurons or astrocytes after oxidative injury. Furthermore, oxidative stress on cortical neurons was reduced when placed in close metabolic contact with SOD-WT overexpressing astrocytes, contrary to cortical neurons placed in close metabolic contact with SOD-mutant astrocytes, which lost their synapsin protein expression after severe glutamate treatment. This observation is a clear indication of the toxicity potentiating effect of the SOD-mutant enzyme.

This microfluidic platform can be used to shed further light on neuron-astrocyte interactions and help expand knowledge on ALS pathogenesis. It can also be used to screen new novel treatment options for ALS and other neurodegenerative disorders.

2.2 Microfluidic Devices in Modelling Cancer

2.2.1 Mimicking Tumour Metastasis

Metastasis can be defined as the spread of cancer to other parts of the body. It generally involves the tumour invading surrounding tissue and entering lymph and blood systems (intravasation) and then traveling to distant tissue. This is followed by an exit from the vascular tissue (extravasation), adapting to the new environment of distant tissue, proliferate, and generate a macroscopic secondary tumour [32]. Although the circulatory pattern of the vascular system plays a role, the cross-talk between specific cancer cell types and receptive environments is substantially significant in deciding the site of extravasation. Breast cancer tends to metastasize to the lung, liver, bone, and brain, while prostate cancer tends to disseminate to bones [33]. A deeper understanding of the metastasis cascade is required to develop new treatment strategies to prevent metastasis and decrease mortality. Current in-vivo and ex-vivo animal models involve using mice and zebrafish embryos; the metastasis process was studied using intravital microscopy [34, 35]. Advanced models involve the intravenous, intracardiac, or direct skeletal injection of breast cancer cells into animals. Current in-vitro models fail in mimicking the in-vivo environment, making results inaccurate from these models.

Bersini [36] developed a 3D microfluidic model to analyse the specificity of human breast cancer metastases to bone. The microfluidic device was designed to have 3 media channels and 4 independent gel channels and was fabricated from PDMS. Osteo-differentiated hBM-MSCs and Human mammary adenocarcinoma cells MDA-MB-231 were used for the experiment. Human mammary adenocarcinoma cells MDA-MB-231 were selected because of their ability to metastasize *in-vitro* and are highly invasive in nature. It is known that cross-talk between MDA-MB-231 and osteo-differentiated hBM-MSCs occurs via inflammatory chemokine CXCL5 secreted by osteoblasts and acting on CXCR2 surface receptors present on breast cells. The extravasation ability of breast cancer cells was quantified based on levels of this selected chemokine.

Cancer cells were allowed to metastasize for 24 h, and extravasation was found in both the test and control. However, the extravasation in the osteo-cell containing chips was significantly higher than collagen gel-only control matrices. Also, the migration distance was higher, and the extravasated cells proliferated to form micro metastases, ranging in size and number of cells. The ability to generate micro metastases demonstrates the ability of this platform to mimic in-vivo events and can be used to understand cancer biology better and help in generating effective anticancer treatment options.

2.2.2 Screening Therapeutic Options for Multi-drug Resistance Cancer

Multidrug-resistant cancer (MDR) tumours protect themselves from the effects of cytotoxic drugs. The overexpression of Pgp transport proteins in their cell membranes is responsible for their resistance, which actively pumps out substrates unidirectionally out of tumour cells through an ATP dependent mechanism, preventing the accumulation of antitumor drugs in the cytoplasm. Antitumor drugs such as anthracyclines (e.g., DOX), vinca alkaloids (e.g., vincristine), podophyllotoxins (e.g., etoposide), and taxanes (e.g., taxol) are rendered ineffective due to this mechanism [37]. Therapeutics that are effective as MDR modulators have not passed clinical trials because of their adverse effects. However, it is hypothesized that combining chemotherapeutic drugs with various MDR modulators can effectively treat MDR. Nevertheless, the number of possible combinations is quite large, making animal testing impractical. Therefore, there is the need for an in-vitro platform that can accurately mimic the in-vivo environment and give valuable insight on all possible pharmacokinetic interactions to determine the efficacy of new treatment strategies.

Tatosian and Shuler [38] developed a microfluidic device that not only analyses the efficacy of combination therapy but also predicts possible adverse reactions. The chip was designed to have four chambers to house four different tissues and was fabricated using Plexiglass. The area of these chambers was decided, keeping in mind several parameters such as cell to total organ volume and cell to liquid ratios. Perfusion rates were adjusted to match organ residence times. The four different types of tissue used were HepG2/C3A representing the liver, MEG-01 cells representing bone marrow, MES-SA, normal cancerous tissue, and MES-SA/DX-5 a multidrug-resistant cancer tissue. Fluorescence analysis was used the check the viability of cells using a combination of CellTracker Green and the live cell stain Celltrace Blue AM.

The chip was then tested using the drugs Doxorubicin (DOX), a commonly used antitumor drug, Nicardipine, and Cyclosporine A(CSP), well documented MDR modulators [37]. First, only DOX was introduced in the system; the C3A cell line showed a significant reduction in cell viability, and the sensitive tumour cells MESSA showed significant cell death. However, the resistant DX5 cancer tissue showed negligible effects. The addition of MDR modulators and DOX significantly increased the cytotoxic response in DX5 cells, while the effects on the other three cell types remained the same. Long term analysis was also carried out by performing this experiment over three days, and similar results were found with DX5 cells being most affected.

There was a synergism seen between the modulators, which were not expected nor seen in previous static culture tests. This information is essential practically, as a single large dose of a single modulator with its side effects can be replaced by combinations of different modulators with varied side effects, ensuring effective treatment and better patient safety. The application of this platform provides an opportunity to play with different drug combinations and their doses and generate reproducible results that are more accurate than current animal models.

Other studies include the modelling of tumour growth in mammary ducts, which can be used to evaluate anticancer therapy options [39]. Sobrino [40] developed in-vitro vascularized micro tumours (VMTs) on a microfluidic device. This 3D culture consisted of human colorectal and breast cancer cells and stromal cells, and a 3D extracellular matrix. The activity of various therapeutic agents such as Vascular Endothelial Growth Factor Receptor (VEGFRs) inhibitors, Platelet-Derived growth Factor Receptor (PDGFR) inhibitors, and Tyrosine-protein kinase receptor (Tie2) inhibitors were analysed, and this platform can be used to study vascularized solid tumours in-vitro.

There exist complex interactions between cancer tumour cells and native body cells. One of these interactions which could shed light on cancer pathogenesis is the interaction between cancer cells and immune cells. The current in-vitro techniques cannot recapitulate these dynamics, while it is difficult to monitor and analyse interactions in animal

models. Businaro [41] developed a microfluidic device using carcinoma and immune cells and successfully simulated previously known in-vivo interactions, and could be used as a tool to investigate further and expand knowledge on these interactions. Kalchman [42] studied the effects of interstitial flow on tumour migration and growth and screened various anti-migratory drugs. In the future, this platform can be used to culture cells obtained from patients to explore patient-specific treatment options. Wang [43] created an oxygen and chemical concentration gradient to understand the dose-dependent antitumor effect of Tirapazamine and investigate oxygen- and chemical-sensitive cell events.

2.3 Microfluidic Devices for Modelling of Cardiovascular Diseases

2.3.1 Lung-On-Chip

Chronic lung diseases such as asthma and chronic obstructive pulmonary disorder (COPD) have been important causes of morbidity and mortality in recent years [44]. They cause airway hyperresponsiveness (AHR), inflammation, and tissue remodelling, often leading to obstruction of airflow, causing patients to experience breathlessness and fatigue [45]. Studies of the pathogenesis of COPD and asthma have revealed that cell–cell communication between Epithelial Cells (ECs) and Smooth Muscle Cells (SMCs) in response to pathogens and other sensitizers is responsible for exacerbations and leading to severe clinical symptoms. However, there remain several questions about these EC-SMC cell interactions. The current in-vitro techniques involve the co-culturing of SMCs and ECs on transwell membrane inserts at the air–liquid interface [46]. However, these inserts are made of rigid, bioinert materials, and the extracellular matrix separating SMCs, and ECs found in-vivo cannot be mimicked by this set-up. Also, physiological air flow exerts a shear force that contributes to the mechanoregulation of ECs. Mice are used as in-vivo models, while lungs obtained from mice have been used as ex-vivo models; however, the biological differences between species hampers translatability of results [47].

Humayun [48] developed a thermoplastic-based microfluidic device that can better mimic the in-vivo environment and allow the study of SMC-EC interactions. The device was designed to have three separate layers fabricated from poly(methylmethacrylate)(PMMA). Human airway epithelial cells were first cultured on the middle layer on a layer of suspending hydrogel, which was then inverted and seeded with human bronchial smooth muscle cells. The epithelial cell facing layer was joined to the top channel while the bottom channel connected to the smooth muscle layer. The bottom layer was filled with media while the top layer remains exposed to air. This device successfully ensured the viability of cell co-culture for 14 days; ECs showed tight junctions in monolayer, while SMCs expressed α-SMA proteins and the cells were highly aligned. All of these findings ensure the cells can perform motility and contractility functions. This platform has potential in studying SMC-EC interactions and also helps uncover how airflow and pollutants affect chronic lung diseases and help better understand their mechanisms.

Stucki [49] tried to mimic organ-level function by recapitulating the diaphragm on their chip. They used flexible PDMS plates to mimic the diaphragm, which was actuated with the help of an electro-pneumatic set-up. Pulmonary alveolar epithelial cells (pHPAEC) were used for this experiment, and they were able to prove that mechanical stress affects epithelial barrier permeability positively. The metabolic activity of these alveolar cells was also found to be higher than in static cultures.

Benam [50] developed a similar device using primary human airway epithelial cells that overlayed on microvascular endothelial cells. The airway epithelium was then exposed to interleukin-13(IL-13), and a decreased ciliary function, hypersecretion of cytokines, and hyperplasia of goblet cells were observed, similar to what is seen in asthmatics. Next, the device was used to culture epithelial cells derived from patients suffering from COPD and analysed, and increased recruitment of neutrophils and cytokine hypersecretion was found. This set-up can be used to identify new biochemical factors playing a role in the progress of the disease and help investigate alternate treatment strategies.

2.3.2 Heart on a Chip

Cardiac side effects is one of the major causes of drug attrition during preclinical drug development [51]. Agarwal [52] designed a microfluidic device consisting of thin film cantilevers made of soft elastomers, referred to as Muscular Thin films (MTFs), which allowed for calculation of systolic and diastolic stresses generated by the muscle tissue. Responses to different concentrations of isoproterenol were recorded. This platform can be utilised for generating large quantities of dependable functional data and can be used to study structure function relationships.

Qian [53] developed a microfluidic device with the ability to measure electrophysiology, contractility and cell adhesion simultaneously on the same device. This integration was achieved using two sensor arrays, Multi-electrode arrays (MEAs) and Interdigitated Electrodes (IDEs). The MEA is responsible for measuring extracellular action potentials IED measured contractions and viability. Human induced pluripotent stem cell-derived cardiomyocytes (Human iPS-CMs) were used for the experiment. The cells were stimulated electrically using a waveform generator generating periodic voltage pulses and the resulting voltage spikes were recorded generating an activation map. Electrochemical impedance

spectra (EIS) were recorded by the IDEs giving vital information about cell growth and adhesion. Drug exposure data was collected by exposing the system to norepinephrine and the results were compared with the existing RTCA CardioECR systems®.

Heart-on-chip technologies are also being used to understand how myocardium's undergo failure to help come up with better therapeutics [54]. They have been used to elucidate the pathophysiology of cardiomyopathy of Barth's syndrome using induced pluripotent cells, helping in understanding the mechanism better and advance research into new treatment options [55].

Microfluidic devices, because of their abilities to prevent the dilution of important biochemical markers that play essential roles in disease pathogenesis, have been gaining an increasing interest as a platform for disease modelling. As various processes involving these chips can be automated and the flexibility of using different analytical techniques, this platform has the aptitude of becoming the drug-testing platform of the future.

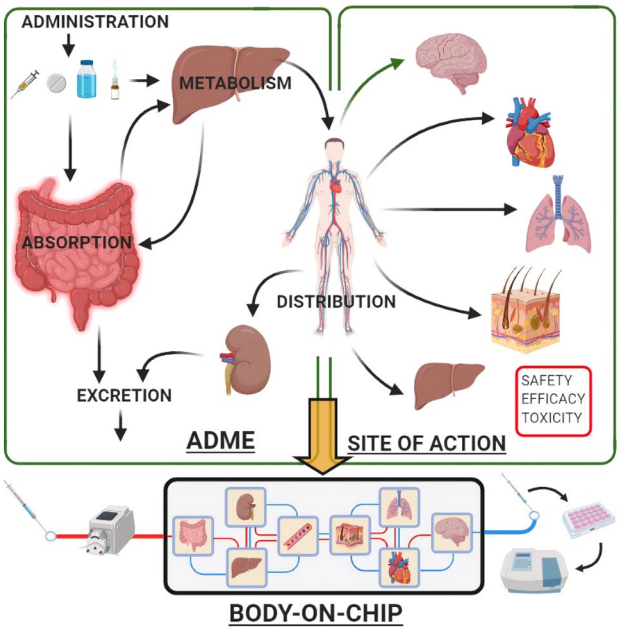


Fig. 2 Overview of PK/PD Modelling and it's possible streamlining using Body-On-Chip System

3 Microfluidic Devices in PK/PD Modelling: Examples of Proof-of-Concept

One of the most significant challenges facing researchers today is determining the toxicity of new drug molecules based on metabolic data generated from current in-vitro and in-vivo animal models. Static in-vitro models fail in mimicking the complex physiological and cell interactions found in a human, leading to failure in predicting the effect of the drug on humans. In-vivo models, although having the ability to mimic the complex physiological environment found in humans, are expensive, labour-intensive, and time-consuming. They also carry the ethical burden of killing a large number of animals to generate the required data. Nevertheless, their predicting capability is low due to the interspecies difference in pharmacokinetic variables. Hence novel in-vitro models are a crucial requirement to streamline the drug development process (as shown in Fig. 2).

3.1 Droplet Microfluidics, a Novel Investigative Tool to Study Drug-Protein Interactions

Pharmacokinetic studies involve the determination of various pharmacokinetic variables such as absorption, distribution, metabolism, and excretion, which in turn determine the efficiency of drugs upon administration to a patient. One crucial interaction playing a role in absorption and distribution is drug-protein binding. Serum albumin is one of the abundant plasma proteins and undergoes non-specific reversible binding with many ligands and drugs [56]. Hence the determination of albumin-binding behaviour is vital in determining ADME properties.

In recent years application of superparamagnetic iron oxide micro- or nano-particles, often referred to as magnetic beads for binding assays, has been getting increased interest [57]. It is a rapid and straightforward method for characterizing ligand-albumin binding [58]. Current liquid handling devices limit the application of magnetic beads due to their inability to study fast kinetics. However, microfluidic-based microreactor devices provide several advantages, including lower costs during early production, lower transportation costs, more manageable to scale up, and higher flexibility to meet market needs [59].

Lombardi and Dittrich [60] demonstrated the effectiveness of microbeads on a microfluidic platform by developing a chip system. In this system, magnetic beads are coated with the target protein, albumin, and formed into microdroplets. The drug under investigation is introduced into the microdroplet. Each microdroplet acts as a microreactor suspended in an immiscible carrier. Microdroplets prevent non-specific interactions with the surface of the microreactor and reduce the volume of reagents required. Radio-labelled C-14 warfarin was used as a model drug for this experiment. Microchannels of the device were designed to include a T-junction and a nickel-coated neodymium magnet placed at one of its arms. The microdroplets on reaching the T-junction split into two droplets of equal volume. The concentration of the drug in the droplet containing the magnetic bead represents the bound drug while, in the other, represents the free drug.

The device was placed under an inverted microscope with a charge-coupled device camera to allow video recording of the process.

To compare the system's effectiveness, a similar experiment was carried out using the conventional bulk-method (as shown in Fig. 3) [58]. After determining tncentrations of free and bound warfarin, the association constant K_A was calculated. The value agreed with reported values, proving microfluidic systems can generate reliable data. Variable channel geometries enable high versatility in droplet generation. The fully continuous operation allows for a variety of applications, such as binding assays, single-cell assays, and kinetic studies, where fast removal of the reactive component is an essential prerequisite.

3.2 Human Blood–Brain Barrier On-a-Chip for High Throughput Screenings

The Blood–Brain Barrier (BBB) is a physical barrier formed by specialized endothelial cells of brain blood capillaries, surrounded by a thick basement membrane. It also shows the presence of other specialized cells, such as astrocytes and pericytes. The BBB has specialized transport systems, one of these transport mechanisms being receptor-mediated transcytosis (RMT), which has been gaining positive interest in therapeutic drug delivery across the BBB without its disruption [61].

Wevers [62] created and maintained a human BBB on a microfluidic platform and evaluated its efficacy. For these experiments commercially available 2-lane and 3-lane OrganoPlate® were employed. Both chips operate without pumps and feature a gravity-driven flow. Brain endothelial cells, pericytes, astrocytes from cell lines originating from human primary cell sources were utilized. These cells were then immortalized using retroviral vectors enclosing an SV40 large T antigen gene that was engineered to promote cell proliferation at 33 °C [63, 64].

The brain endothelial cells were cultured using the 2-lane OrganoPlate while the BBB co-culture was prepared using a 3-lane OrganoPlate®. Barrier integrity was analysed by introducing FITC-dextran, a fluorescent probe utilized to carry out permeability studies, in the top channel of both the chips. The leaking of the fluorescent molecule across the membrane was monitored. The ratio of the fluorescent signal across the membrane was then analysed to understand membrane integrity. Immunostaining was also carried to check for the expression of essential adhesion molecules.

TY10 brain endothelial cells grown in the 2-lane plates on staining with ActinRed showed vessel-like structures of endothelial cells. Immunostaining confirmed the expression of molecules such as vascular endothelial cadherin (VE-cadherin), endothelial cell adhesion molecule 1(PECAM-1), and claudin-5. The presence of these markers indicates successful barrier formation. Barrier integrity test revealed complete retainment of the fluorescent dye. These results were compared to epithelial cells cultured under static conditions, and more extensive expression of vital junctional proteins was observed in the microfluidic device culture. Cells cultured under perfusion showed tighter barrier formation than cells cultured under static conditions. The BBB culture in the 3-lane OrganoPlate® showed similar results to the barrier integrity test.

Next, antibody transcytosis across the BBB cultures in the 3-lane OrganoPlate® was analysed. For this purpose, two antibodies, Anti-human transferrin receptor mouse monoclonal antibody MEM-189 mIgG1 and anti-hen eqq lysozyme (anti-HEL) antibody, were employed. Positive results were obtained, demonstrating this platform's effectiveness in being a useful tool for screening of drugs targeting the brain.

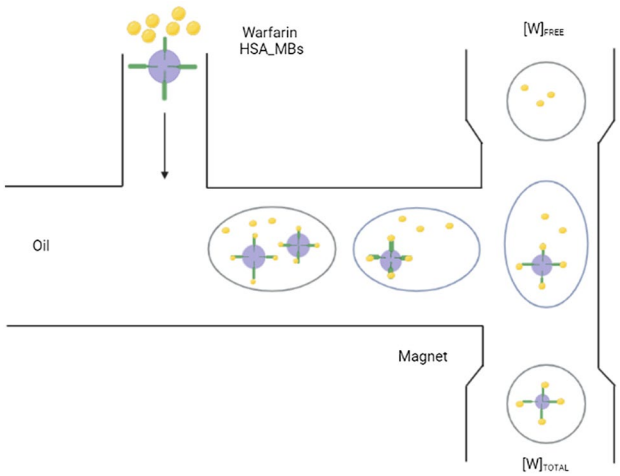


Fig. 3 C-14 warfarin representation in droplet microfluidic

3.3 Gut-On-Chip in Predicting Drug Permeability

The current in-vitro assay procedure for determining drug permeability involves the culturing of Caco-2 cells, derived from human colon adenocarcinoma, in a transwell system. The molecule to be analysed is introduced into the well and allowed to permeate freely across the cell monolayer. However, this assay procedure has several drawbacks: requiring a large number of cells and other reagents, long testing times, and also their inadequacy in mimicking the dynamic environment of human intestines.

Gao [65] designed a microfluidic platform integrated with Electrospray ionization quadrupole time of flight mass spectrometry (ESI-Q-TOF MS). A micro solid-phase extraction (SPE) channel integrated with the microfluidic device was used for sample pre-treatment before MS detection. Curcumin was chosen as the model drug to evaluate the performance of the system.

The microfluidic device consisted of two separate plates with suitable microchannels, one plate forming the upper channel and the other the lower channel. A polycarbonate membrane (PC) modified using two reagents, poly-L-lysine and Matrigel, was sandwiched between the two PDMS layers, a modification essential for cell adhesion. A predetermined number of Caco-2 cells were introduced through the upper channel, and a pipette attached to the outlet was used to generate negative pressure, allowing for cells to fill the channel and prevent accumulation at the inlets and outlets. Curcumin was then introduced into the chip and the curcumin that permeated to the bottom channel was then passed through an SPE column, a microfluidic device filled with C-18, integrated on-chip to desalt and concentrate curcumin before analysis.

Permeation values were determined using ESI-Q-TOF MS which matched with values in literature. Cell viability was evaluated at the end of the experiment and integrity of the cell monolayer was evaluated by exposing the cells to Lucifer Yellow (LY). LY is impermeable to Caco-2 cells, and the monolayer was exposed to this agent for 1 h. Less than 1% of permeation was observed, demonstrating the integrity of the cell monolayer. This experiment shows the viability and flexibility of this platform for carrying out permeability assays. The microfluidic platform shows the potential to be used as a useful tool for drug screening in the early stages of drug development.

Kim [66] developed a biomimetic "human-gut-on-chip" which not only mimics the structural, functional and pathophysiological functions of the human gut but also account for crucial microbial symbionts that are found *in-vivo*. Lactobacillus rhamnosus GG, a microbe found in human intestines was co-cultured along with intestinal epithelial (Caco-2) cells. Such a platform can be used to assess drug absorption of new drug candidates and can also be used to model intestinal diseases.

3.4 Liver-On-Chip in Predicting Hepatic Clearance

Chao [67] employed a microfluidic HμREL® chip to determine the hepatic clearance of six marketed drugs (carbamazepine, caffeine, timolol, sildenafil, imipramine, and buspirone) in humans. It was used to evaluate pharmacokinetic and pharmacodynamic interactions between the drug and cultured cells under perfusion. For this evaluation, four microfluidic HμREL® chip were connected in parallel. PEEKsil™ tubing was used to reduce non-specific binding. A peristaltic pump was employed to maintain the flow of fluid through the chips. Outflow from the chips was collected in glass vials, which acted as reservoirs and de-bubblers.

Cryopreserved hepatocytes were used for this experiment which were seeded into the liver compartment and then allowed to attach to the biochip surface in an incubator. The drugs were introduced into the chips by replacing the medium in the reservoirs with a drug-containing medium. Aliquots were drawn at predetermined time points and analysed by LC–MS/MS. A static culture was prepared in a 96-well titre plate and run in parallel to compare performance with the microfluidic system.

The calculated values of hepatic clearance from the experiment agreed with the reported values demonstrating the efficacy and accuracy of the microfluidic platform. Also, this experiment demonstrates the advantage of perfusion. Not only is it favourable in maintaining cell viability and morphology, but it also ensures metabolic activity for a more extended period. It also presents itself as a useful tool for investigating drug-drug and other interactions with tissue underflow.

Several experiments are being carried out to improve the microfluidic platform. Although the closed and compact nature of microfluidic devices offers several advantages, there are some pitfalls associated with this too. The major drawback is difficulty in analysis and detection in a non-invasive manner. Various popular analytical methods, such as fluorescence detection, are often used for evaluating cell viability, metabolic activity, protein expression, and oxygen concentration [68, 69]. However, these conventional methods are not optimized for perfused microfluidic devices. The tubes required for perfusion in a microfluidic device would have to be disconnected for evaluation under a fluorescence plate reader. Repeated disconnecting and reconnecting hinder the continuous operation of the device and also increases the chances of a bacterial infection. Air bubbles could also be introduced in the process, and overall real-time analysis is not possible. These obstacles have resulted in microfluidic devices undergoing end-point measurements.

Sung [70] developed an In-Situ Fluorescence Optical Detection System (ISFODS) for microfluidic devices based on Ethoxyresorufin-O-dealkylase (EROD) assay. The Ethoxyresorufin-O-dealkylase (EROD) assay is widely used to quantify P450 1A1 and P450 1A2 enzyme activity and involves the measurement of conversion of ethoxyresofurin to resofurin, a fluorescent product [71]. The ISFODS consists of a green light from a LED with a known peak wavelength and is used to excite resofurin in the detection chamber. The resulting fluorescence emissions are focused with a lens and detected with a low-power photodiode. The significant advantage of this system is the use of a single photodetector. HepG2/C3A cells were cultured in a micro cell culture analogue (μCCA), a microfluidic device designed to sustain multiple cell cultures to test the system. The μCCA system integrated with ISFODS was tested with different concentrations of substrate and different cell densities. Higher substrate concentrations lead to higher conversion kinetics. Similar observations were seen upon increasing cell density. These experiments demonstrate the effectiveness of

the ISFODS system and can provide real-time data in-situ. Several μCCA devices can be run in parallel coupled with this system which can help in high-throughput screening.

3.5 Liver-On-Chip in Predicting Cytotoxicity

Mao [72] developed an integrated microfluidic platform that imitates drug metabolism. Electrospray ionization quadrupole time of flight mass spectrometry (ESI-Q-TOF MS) was used to detect drug metabolites, with sample pre-treatment carried out using a fused silica capillary integrated with the chip. Acetaminophen (AP) was used to validate the effectiveness of this platform, as it was found to be safe in animal experiments but lead to liver failure in humans, demonstrating the shortcomings of current toxicity testing methods [73]. The microfluidic device was designed to have three functional sites for metabolism, cytotoxicity testing, and a pre-treatment chamber for sample pre-treatment. For metabolism, the devices were designed to have zigzag channels where Human Liver Microsomes (HLMs) encapsulated with PEG were placed to form a bioreactor. For cytotoxicity assay and metabolite analysis, the devices were designed to have straight channels. Hepatoma G2 (HepG2) cells were used to determine cytotoxicity, seeded into the straight channels of the microfluidic device. The sample drug was then introduced into the reaction chamber and was incubated for 12 h.

Part of the resulting metabolites was passed into the SPE channel for pre-treatment, while the rest were introduced into the toxicity testing chamber. The cells were exposed to metabolites for 12 h. Cell viability was analysed using fluorescence analysis. Different sample mixtures were prepared and cell viability was determined for each. AP and APG levels were determined with ESI-Q-TOF MS, and the AP concentrations could be correlated to cell toxicity. This experiment successfully demonstrates this platform's effectiveness to imitate liver metabolism and investigate cytotoxicity and act as a useful tool in metabolite screening.

Kostadinova [74] developed a microfluidic device to predict Drug Induced Liver Injury (DILI) by establishing a three dimensional (3D) liver co culture system with primary parenchymal and non-parenchymal hepatic cells along with hepatocytes, stellate, Kupffer and endothelial cells. The most important feature is the ability of the system to maintain liver function up to three months. The platform was then tested using a range of known hepatotoxicants and more accurate results were generated than current monolayer hepatocyte cultures.

3.6 Liver-Tumour-On-Chip

The use of 3D cell cultures allows for better elucidation of cell–cell interactions in human tumours and ensures that the data generated is more reliable than conventional 2D in-vitro cell cultures [75]. The use of biomaterials such as gelatine methacryloyl (GelMA) hydrogels allows the replicating of mechanical structures of the native extracellular matrix (ECM). They encapsulate cells and form a 3D scaffold and closely mimic the mechanical properties of ECM [76]. However, they lack essential growth factors to promote cell growth and maintain cellular functions. Hence it becomes imperative to substitute these factors externally.

Lu [77] overcame these limitations by developing a biomimetic 3D liver on-chip using GelMA integrated with a decellularized liver matrix (DLM). The DLM helped provide essential components, such as growth factors and scaffold proteins, to mimic the native organ more effectively. GelMA when mixed with DLM solution and a suitable photo-initiator, on exposure to UV light undergo cross-linking, changing the solution state to the solid-state. Hepatocytes were encapsulated by the addition of HepG2 cells to the DLM-GelMA solution, followed by UV light exposure. Observing the resultant DLM-GelMA biomaterial under a scanning electron microscope revealed a porous microstructure and an enhanced smoothness of GelMA hydrogels which can

Table 1 Overview of recent developments using organ-on-chip platform

Sr. no.	Description	References
1	This research aimed to mimic the progression of kidney cancer in the liver. For this purpose, a new metastasis-on-a-chip model was designed in which kidney cancer cells (Caki-1) and hepatocytes (HepLL) were cocultured. The platform was then used to assess therapy strategies	[100]
2	Here we see the development of a pancreas-liver coculture in a microfluidic device. Rat Islet of Langerhans and hepatocytes were used. The viability and function of the cells was compared with static monocultures. The device was used for organ-to-organ interaction studies	[101]
3	In this research article we see the development of a organ-on-chip supporting liver, vascular, gastrointestinal and kidney tissues. 96 of such devices were then combined into an industry standard plate and used for carrying out high-throughput screening	[102]
4	This research was involved in developing a foetal-membrane-organ-on-chip device (FM-OO-C). The device consisted of two chambers and Primary AECs and decidual cells were cultured. The effect of cigarette smoke on these cells was analysed	[103]

be attributed to the adhering of ECM proteins and growth factors on the surface (Table 1).

The microfluidic device was fabricated from polymethyl methacrylate (PMMA) with polyethylene terephthalate (PET) and predetermined cell densities of HepG2cells mixed with the DLM-GelMA solution. The resultant solution was then pipetted into the microfluidic device's chambers. The viability and functionality of the 3D cell culture were monitored by measuring ALB and urea production and live/dead staining. Long term analysis of the 3D cell culture over 21 days revealed a higher secretion of urea and albumin in comparison to the GelMA-only control. These results demonstrate the superiority of DLM-GelMA in supporting hepatocyte growth and function.

The biomimetic liver-tumour-on-chip was then utilized to generate drug-response curves. Drugs acetaminophen and sorafenib were chosen. The analysis was carried out at different concentrations, with the exposure period being 24 h. Linear dose-dependent drug response was observed for both the model drugs in cell viability, urea, and albumin production. Significant cell death occurred upon increasing the concentration of both drugs.

The integration of structural proteins and liver-specific growth factors, along with the biomimicry of 3D liver microenvironment using scaffolds, ensures accurate simulation of the environment that will exist in a human. Hence it presents itself as a better in-vitro disease modelling alternative increasing the efficiency of pharmacological studies.

Zuchowska [78] developed a microfluidic device for the high-throughput assessment of cytotoxicity of anti-cancer drugs on liver cancer cells. HepG2 spheroids were used for this purpose and 5-fluorouracil (5-FU) was used as the model drug. The chip was designed to have three rows of microchambers which are connected via microchannels. Each microchamber consisted of 18 identical microwells semi spherical in shape to promote self-aggregation rather than aggregation to chip surface. On exposure to the drug, a reduction in size of the spheroids was observed, however chronic exposure resulted in resistance to the drug with the spheroids recovering in size. Liver-On-Chip technologies provide higher accuracy, sensitivity and can be an important tool in future research endeavours.

3.7 Kidney-on-a-Chip

Nephrotoxicity causing kidney damage is an important cause of drug failure. In the preclinical stage, it only accounts for 2% of cases; however, it becomes more severe in Phase 3 and hospital and community settings, accounting for 19% and 20% cases, respectively [11, 79]. This sudden increase in affected cases demonstrates the ineffectiveness of current preclinical cell culture models which can be attributed to the lack of typical microenvironment cues found in vivo that promote histodifferentiation. Kidney-on-chip presents an opportunity in reconstituting relevant physiological microenvironments enabling cells to express a more comparable in-vivo phenotype and ensuring more human-like responses.

Jang [80] designed a polydimethylsiloxane (PDMS) microfluidic device consisting of two flow channels separated by a porous polyester membrane. Primary human kidney proximal tubular cells grown in a suitable growth medium were seeded onto the upper layer of the porous membrane under static conditions and incubated to reach confluency. Control was set up by seeding cells on an ECM treated porous polyester membrane placed within inserts of a Transwell plate. Fluid shear stress was applied through one flow channel of the microfluidic device forming the "luminal" flow channel while the other flow channel was kept static, forming the "interstitial" compartment. The applied shear stress caused the cells to return to their typical columnar form and increase their height by 200%. Normal kidney epithelial cell polarity was established, indicated by Na/K-ATPase being restricted to the basolateral membrane. A 200% increase was observed in the expression of NA/K-ATPase and aquaporin 1(AQP1) by 300%.The expression of these cross epithelial transporters is physiologically relevant as they are crucial in the functioning of the renal tubules [81]. Exposure to shear also promoted the expression of primary cilia essential for maintaining tubular morphology, and this expression was significantly higher than cells cultured under static conditions [82, 83].

To ensure the tubule-on-chip carried out physiological transport functions, the uptake of fluorescently-labelled albumin was studied. A two-fold increase in absorption was found in comparison to the control. There was also an increase in the expression of critical brush-border enzymes, such as enzyme alkaline phosphatase (ALP). An increase in sodium-glucose cotransporter 2 (SGLT2) was seen, leading to increased reabsorption of glucose by secondary active transport compared to the static Transwell system. These findings point to a more “in-vivo” phenotype being expressed by cells in a microfluidic device.

Next drug-induced cell toxicity was modelled using the chip. For this purpose, a chemotherapeutic drug and a known nephrotoxin was chosen, Cisplatin. Cisplatin tends to accumulate in the renal parenchyma leading to the generation of reactive oxygen species (ROS), causing cell damage. Cell damage was tracked by measuring lactate dehydrogenase (LDH) release while cell apoptosis was determined using TUNEL assay or Annexin V staining. Cells under perfusion showed less cell damage and apoptosis in comparison to static Transwell culture. Moreover, the recovery of cells post-exposure to Cisplatin was significantly higher for cells under perfusion.

To conclude the study, the activity of P-glycoprotein ATP-binding cassette membrane transporter (Pgp) was

analysed as they are known to mediate efflux of certain chemotherapeutic drugs, which may lead to multidrug resistance [84]. The activity of this transporter was analysed using calcein-acetoxymethyl-ester (calcein-AM), which is either effluxed by Pgp or hydrolyzed into fluorescent calcein, leading to an overall decrease in fluorescence. Inhibition of this transporter would result in the accumulation of calcein in the tubular cells. Pgp inhibitor verapamil was chosen, and the intracellular fluorescence was lower in the microfluidic device than in the static culture. These results demonstrate the effectiveness in the measurement of Pgp activity on this platform, which is essential for anticipating drug toxicities and drug-drug interactions.

Kimi [85] developed a microfluidic kidney model which was then used to shed light on the difference in drug reactions to different pharmacokinetic regimens. Gentamicin was used as the test drug for this experiment. The chip consisted of two layers made of PDMS with a porous polyester membrane placed between the layers. Madin-Darby canine kidney (MDCK) cells were seeded onto the porous membrane followed by coating with fibronectin. Shear was applied to induce histodifferentiation in the cells. Two dosage regimens were devised duplicating the pharmacokinetics of human bolus injection and continuous infusion in humans. A decreased junctional protein immunoreactivity was observed in the case of both dosage regimens, with a disruption in cell junctions and increased permeability. However, the nephrotoxicity in the case of bolus injection regimen was found to be less significant than continuous infusion.

These findings demonstrate the effectiveness of the microfluidic platform in simulating in-vivo microenvironments to generate more reliable results. It is a cost-effective tool that also enables direct visualization and quantitative analysis of various mechanisms within renal tubules not possible in traditional animal models.

3.8 Co-cultures in Microfluidic Devices

In this section we will be taking a look at microfluidic devices integrating two different tissue types to understand effects of drugs on two different systems at the same time. Microfluidic systems can be designed to sustain different types of cell tissue simultaneously and provide a unique opportunity for the simultaneous analysis of effects of drug candidates.

3.8.1 Liver-Immune Co-culture for Predicting Skin Sensitization

Drug-induced skin sensitization is prevalent worldwide and is the third leading cause of non-fatal disease burden globally, most commonly seen in drugs administered via transdermal, oral, or intravenous routes [86]. However, there exists a lack of in-vitro models capable of predicting drug-induced skin sensitization with adequate accuracy [87]. This lack arises from the fact that drug-induced skin sensitization follows more complex pathways in comparison to contact dermatitis. Often the drug molecule itself does not precipitate skin sensitization but rather its metabolites generated by the liver. It is postulated that these reactive metabolites are taken up by antigen-presenting cells (APCs), processed, and subsequently presented to T-cells leading to an immune response [88]. A skin-centric in-vitro model is rendered inadequate as they are unable to incorporate all cellular mechanisms, which lead to skin sensitization. Thus, an in-vitro model able to mimic the generation of drug metabolites and the immune cascade's activation could be a powerful tool in measuring the skin-sensitization propensity of newer drugs.

Keeping these elements in mind, Chong [89] developed a microfluidic device that incorporates human progenitor cells (HepaRG)-derived hepatocytes and U937 immune cells. HepaRG cell line expresses genes encoding several liver proteins, including CYP-enzymes and transporters involved in Phase-2 metabolism [90] and hence can mimic hepatocyte structure as well as function. The U937 cells can differentiate between contact-sensitizers and non-sensitizers [91]. The significant advantage of using a microfluidic platform is the small volume of medium required for operation ensuring metabolites key in eliciting an immune response do not get diluted. The dilution of metabolites is a significant disadvantage of current bulk cultures, often leading to weak or no immune reactions and false conclusions. The compartmentalized design of microfluidic devices allows for the preservation of the native tissue microenvironments and allowing cross-talk between the cells at the same time.

The chip was designed to ensure the uniform diffusion of metabolites into the immune compartment. Predetermined cell densities of HepaRG and U937 were introduced along with suitable culture mediums. A conventional Transwell system was also set-up to compare the difference in performance.

Drugs carbamazepine, phenytoin, and allopurinol were chosen to analyse the predictive capability of the chip based on reports of them leading to severe cutaneous drug reactions. Acetaminophen was chosen as the control drug as it very rarely causes skin sensitization. The cell cultures were exposed for over 48 h. Analysis of the metabolites was carried out using Liquid chromatography–mass spectrometry (LC–MS) and metabolites of the drugs which precipitate skin sensitivity reactions were looked for. The detection of all reported metabolites assured the performance of HepaRG cell culture. Next, the response of the immune cells was analysed using U937, a myeloid cell line that changes from monocytes to APCs on exposure to antigens. This change is characterized by elevation of levels of IL8, IL1β, CD86 [91].

This is the basis of characterization with the U-SENS assay in OECD guidelines and enables differentiation between transdermal skin sensitizers and non-sensitizers.

The activation of U937 cells was recorded and was found to be significantly higher than the Transwell co-culture system. This difference was significant for phenytoin and allopurinol. The examination of cell surface cultures of the microfluidic device revealed the expression of co-stimulatory receptor CD86. Only carbamazepine induced upregulation of CD86 in the Transwell co-culture, while the other two drugs did not. These findings demonstrate the higher consistency and accuracy of the microfluidic platform to conventional systems. These advantages are due to the lower rate of diffusion between two tissue culture compartments, effective generation of reactive metabolites, and low volumes of media required for functioning, proving to be a novel screening platform to identify drugs that could lead to skin sensitizations.

3.8.2 Gut-Liver Co-culture in PK Evaluations

Here we see the integration of liver and gut on a single chip allowing analysis of absorption and metabolism, two critical pharmacokinetic variables deciding half-lives and drug toxicities.

Tsamandouras [92] designed a multi-micro physiological system (MPS) fabricated using polysulfone plastic to minimize drug absorption. The chip was designed to have five distinct compartments, one of which acts as a mixing chamber, interconnected via microfluidic channels with peristaltic micropumps for perfusion. The mixing chamber aims to mimic the systemic circulation of the body. A typical transwell-style Gut MPS was created using Caco2-BBE epithelial cells and mucin-producing goblet cells, along with an immune element consisting of primary monocyte-derived dendritic cells. Liver MPS was prepared using scaffolds to give a 3D structure using primary cryopreserved hepatocytes and Kupffer cells. The integration of the immune component is to mimic in-vivo micro physiology better and to enable future applications in pharmacological studies. The fluid flow through the platform was controlled to mimic flow partition in-vivo. The flow from the mixing chamber was partitioned as 75% flowing to the gut compartment while the rest 25% flowed to the liver compartment. While the gut compartment receives the entire input flow from the mixing chamber, the liver compartment received $1/4^{th}$ of input flow from the mixing chamber and the rest from the gut compartment representing hepatic artery flow and portal vein blood flow, respectively. The design of the chip also allows for flexibility in dosing. If oral administration is to be mimicked, the drug can be introduced directly on the gut compartment's apical side, allowing adsorption followed by transport to the liver compartment. If IV administration has to be mimicked, it can be directly introduced into the mixing chamber.

The performance of the platform was evaluated by determining the kinetics of two compounds Diclofenac and Hydrocortisone. Diclofenac was dosed to the gut MPS' apical side, and the concentration–time gradient was obtained by regular sampling from the mixing chamber. A plasma concentration profile, similar to oral administration in humans, was observed. Similar results were found in the case of hydrocortisone. This experiment demonstrates the platform's potential for determining several PK variables like drug absorption, drug-drug interactions, accumulation of drug metabolites, and effects of repeated administration of a drug.

3.8.3 Liver Kidney Co-culture in Assessing Toxic Metabolites

Hepatotoxicity and nephrotoxicity are important causes of drug candidate failure. Liver Kidney co-cultures can be used to study liver-associated kidney toxicity.

Theobald [93] designed a microfluidic device consisting of three inlet ports and three outlet ports and having two interconnected chambers, the liver and kidney compartments, of equal geometric dimensions. The connection allows for the flow of medium and drug metabolites between the liver and kidney compartments. HegG2 and Hek293 cells were seeded into the liver and kidney compartments respectively and incubated under static conditions in a humidified incubator before exposure to perfusion.

To compare results 96 well plate monocultures were also prepared that underwent exposure to the same agents. Various assays and evaluation techniques were used to evaluate the chip. Cell viability of HepG2 cells was evaluated using enzyme-linked immunosorbent assay (ELISA) for albumin production, urea production using the urea assay kit and gene expression of CYP450 family genes. There was a significant decrease in albumin production and urea synthesis in the static culture while remaining constant in the microfluidic system. Gene expression of all 6 CYP450 family genes was significantly higher than the static culture, suggesting the microfluidic system is more suitable than static cultures. Hek293 cells were analysed based on FITC-albumin uptake, and constant flow promoted higher uptake than static culture.

Next the platform's performance was evaluated using Aflatoxin B1(AFB1) and Bezoalphapyrene (BαP). AFB1 introduction in the chip with the fluid flow direction from the liver compartment to the kidney compartment leads to lowered cell viability by 20% in both the compartments. However, when the flow was reversed, the viability of Hek293 cells was not significantly lowered and can be linked to the non-conversion of AFBI to its toxic epoxide by liver cells, confirming that metabolism of AFB1 leads to its significant

side effects. Next, Rifampicin, a potent inducer of CYP3A4, was co-incubated with AFB1. Western blot analysis revealed a two-fold induction in HepG2 cells in 24 h while Hek293 cells remained uninduced. Rifampicin did not affect the toxicity of AFB1 in the static culture, but the expected synergistic effect was seen in the microfluidic culture. Similar results were observed with the administration of BαP.

These results provide encouraging evidence about this platform's capability and scope to create multi-organ chips and revolutionize in-vitro testing.

3.9 Multi-organ on a Chip

In this section, we will be discussing examples of microfluidic chips incorporating multiple tissue types. They represent the final goal of the platform to be a comprehensive testing tool allowing analysis of effects of drugs on multiple systems simultaneously.

Oleaga [94] developed a microfluidic device with a gravity-driven flow system, avoiding the use of peristaltic or other types of pumps, tubing, and bulky valves, lowering costs. It also solves the problem of air bubbles entering the system which can be detrimental to the tissues. The microfluidic device was then used to culture viable human liver, cardiac, skeletal muscle, and neuronal cell cultures. These cell cultures were chosen to shed light on the metabolic and functional changes that occur on exposure to various compounds. For this study, compounds with established toxicological properties were selected. The viability and functionality of these cell cultures were demonstrated over 14 days, making it an effective tool for evaluating chronic effects. The analysis of the cell cultures was carried out in non-invasive in-situ form and ex-situ after complete disassembly of the chip, allowing for extensive testing. Compounds Doxorubicin (DOX), Atorvastatin (ATR), Valproic acid (VPA), and Acetaminophen were used to evaluate the chip. The control compound for the experiment was N-Acetyl-m-aminophenol (AMAP).

DOX is known for initiating apoptotic pathways and cause increased production of Reactive Oxygen Species (ROS). The primary site of metabolism of DOX in the human body is the liver, generating high amounts of oxidative stress in the organ leading to hepatotoxicity, a well-known side effect of DOX [95]. Introduction of DOX into the system led to a loss in cell viability of cardiomyocytes and hepatocytes. Electrophysiology of the neuronal cell culture remained unaffected, with parameters such as resting membrane potential and peak ionic currents showing no difference with the untreated control. However, there was a loss in cells and repetitive and spontaneous firing ability.

Atorvastatin has a rare but significant adverse effect of causing severe muscle damage (rhabdomyolysis) [96]. The introduction of ATR in the system showed a decrease in the viability of hepatocytes and skeletal muscle cells. An increase in urea production occurred while there was no change in albumin production. In cardiac tissue samples, there was an increase in cell viability as well as functionality. Cell loss in neuronal cell culture occurred within 48 h of introduction in the system.

Valproic Acid (VPA) leading to liver failure is reported; however, the incidence rate is very low. However, the rates are significantly higher in children under the age of 2 years [97]. On the introduction of cytotoxic concentrations of VPA in the four organ systems, a decrease in hepatocyte viability occurred. A decrease in urea production indicated that hepatic metabolic function was slightly affected; however, no changes occurred in HSA levels. An increase in cardiomyocyte viability and the beat frequency was recorded, demonstrating the cardioprotective effects of VPA. VPA showed no effects on skeletal muscle cell viability or contractile activity. Analysis of neuronal electrophysiology remained similar to that of the control.

Upon the introduction of APAP in the system, a decrease in liver cell viability occurred. There was an increase in urine production, but albumin production remained unaffected. The viability of cardiomyocytes and skeletal muscle cells remained unaffected. However, a reduction in cell numbers of skeletal tissue culture that contracted occurred. There was an increase in the beating frequency of cardiomyocytes. The study of electrophysiology of neurons revealed a depletion in repetitive and spontaneous firing ability. However, stimulated action potentials and ion currents remain unchanged, demonstrating adequate neuron health. However, 48 h exposure to APAP leads to decreased neuronal cell numbers, contradicting the literature.

N-acetyl-m-aminophenol (AMAP) has an antipyretic and anti-analgesic activity similar to APAP. However, it does not lead to hepatotoxicity and hence is often used as a negative control to study the toxic effects of APAP. AMAP introduction in the system caused no changes in the viability of hepatocytes and neuronal cells. Neuronal electrophysiology remained similar to the control, even after 48 h of exposure. Skeletal muscle cell viability and contractility also remained unaffected. However, there were adverse effects on cardiomyocyte cell viability, and there was a drop in beat frequency. AMAP was slightly cardiotoxic in this 4-organ set-up.

The above study establishes a four-organ system with continuous recirculation of serum-free medium. The system is modular and easily reconfigurable. Liver, cardiac, neuronal, and muscle cultures were used in the above experiment as they represent critical organ systems that experience significant side effects, leading to drug candidate failure. The measurement of functional parameters such as contractility, electrical and mechanical responses, and metabolic biomarkers allows for a more detailed understanding of physiological responses to

drugs. The system can maintain the viability of cells for over 14 days allowing for the study of five different drugs in this period.

Sung and Shuler [98] developed a microfluidic device to maintain different tissue types which was used to investigate the cytotoxic effects of anticancer drugs. The μCCA was designed to have three chambers to hold colon cancer, liver and marrow cells respectively, interconnected by microchannels mimicking the blood flow pattern in-vivo. Colon cancer cell line (HCT-116), hepatoma cell line (HepG2/C3A) and Myeloblast cell line (Kasumi-1) were used in the colon cancer, liver and marrow chambers respectively. All cells were embedded in 3D hydrogels as it provides a more accurate physiological environment than a monolayer culture. Tegafur, an oral prodrug of 5-fluorouracil (5-FU) was chosen as the test drug, while cells cultured in 96-well microtiter plate were used as control. Tegafur was metabolised to 5-FU by liver cells in the μCCA device while cells in the 96-well microtiter plate were unable to. Hepatotoxic, haematological toxicity and tumour killing effects of the drug were recorded, which agreed with published clinical data.

Li [99] developed a multilayer microfluidic device to maintain 4 different cell types, liver, breast cancer, lung cancer and gastric cells. The chip was designed to have two layers with a porous polycarbonate membrane sandwiched between the layers. The upper layer consisted of a single chamber containing liver cells forming the liver chamber while the rest of the cell types were contained in the bottom layer in three different chambers. The porous membrane allowed for the movement of drug and its metabolites freely between the tissues. Capecitabine (CAP), a chemotherapeutic was chosen as the test drug. CAP was metabolised by the liver cells to its active metabolite 5-Fluorouracil (5-FU). A metabolism-dependent effect of the drugs on the tissues was observed, comparable to the effects seen *in-vivo*.

The various applications discussed in this section demonstrate the potential of microfluidic devices to be powerful tools in improving predictions during preclinical assessments of drug candidates, reducing chances of failure, and increasing the health industry's efficiency. Important pharmacokinetic and pharmacodynamic variables can be studied simultaneously with these platforms with results generated being accurate and reliable, reducing the chances of drug failure later in the drug development process. They can be utilized in the development of personalized medicine and could help in improving the safety profiles of pharmaceuticals.

4 Future Prospects of Microfluidic Devices

4.1 Application in Personalized Medicine

Personalized medicine is a new and upcoming branch in health care that aims to diagnose and treat patients with diagnostic tools, medicines, and dosage regimens tailored to each patient. This approach takes into consideration the unique characteristics of the patient and his/her disease. The growing knowledge about genetic polymorphisms across populations introduces variance in metabolizing enzymes, target proteins and transporter proteins. Fluctuations in these critical parameters often leads to varied efficacy and severe side effects of drugs. The medical industry currently is undergoing a shift from population-centric to patient-centric, providing safer and efficacious treatment options.

Microfluidic systems or Organ-on-Chips are proving to be useful tools in aiding this paradigm shift in the healthcare industry. Microfluidic platforms could be populated with one's cells, creating a "You-On-chip" that opens up several avenues (as shown in Fig. 4). They can be used to generate critical information about disease mechanisms, drug toxicities, their efficacy, and metabolism profiles. These factors could prove critical in deciding the dosage regimen and deciding whether a certain treatment strategy is suitable for the patient or if alternate treatment strategies could prove beneficial. It can also help uncover unexpected side effects improving patient safety and deleting physician guesswork. This could also potentially lead to "first-in-human" clinical trials where drug candidates could be tested for their efficacy

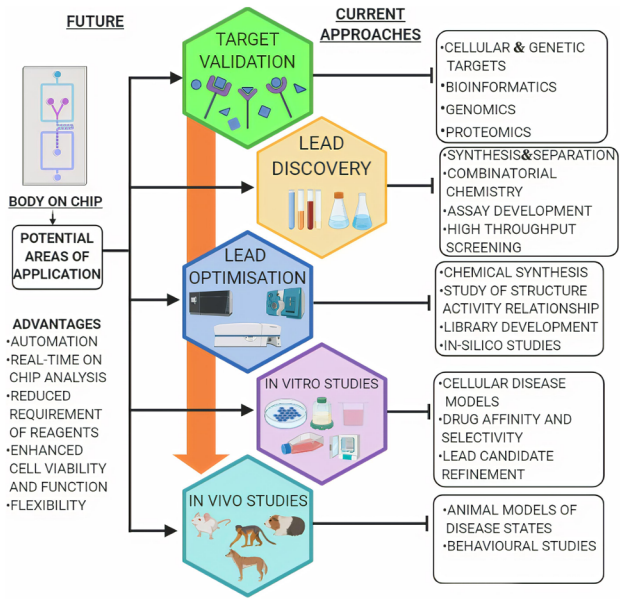


Fig. 4 Future of Drug Discovery with Organ-On-A-Chip System

and safety without requiring any volunteers. The healthcare industry currently follows a "one size fits all" approach, which often is detrimental in rare diseases where familial genetic mutations often lead to significant fluctuation in the efficacy and safety of experimental drugs. It is often difficult for these patients to participate in clinical trials as they may be unable to reach hospitals where these trials are being conducted or may be unable to participate in a trial as they may already have participated in other trials in the past. In such cases, cells from the patient could be used to perform a clinical-trial-on-chip to understand the efficacy and toxicity profile of the experimental drug before administration to the patient. Appropriate amounts of information are at times unavailable for rare diseases. Different tissues obtained from an affected patient could help model the disease on a chip and shed light on previously unknown facts. These discoveries could lead to allowing better testing of a variety of agents without harming the patient.

Several diseases are associated with genetic risk factors; hence chips using individual patient cells can help map these pharmacogenomic variations and help identify population subgroups that may be susceptible to adverse reactions from a particular treatment strategy. It may also help identify populations that may be more effectively treated. Advances made in gene editing tools such as CRISPR-Cas9 open up possibilities to remove or add genes to simulate pathological conditions arising from specific genetic abnormalities. It may shed further light on the genetic basis of disease and allow isolation of genotype and phenotype precisely for the disease [104]. Microfluidic systems could also shed light on previously unknown factors and be revolutionary for fields such as genomics, transcriptomics, proteomics, and many others.

Cancer is a unique disease state where tumorous cells use the body's proliferative processes and can hide from the body's immune system, and it has been extensively researched with no definitive cure in sight. There has been increased interest in immunotherapies, which aims to use the body's immune cells to mount an attack on the cancerous cells. There have been recent approvals by the US FDA for such treatments, like Kymriah and Yescarta. Although offering several advantages over current chemotherapy and radiotherapy treatment strategies, they are not without serious side effects that might seriously harm patients' lives and is an expensive form of therapy. Microfluidic systems could prove helpful in the treatment of cancer patients. Biopsy tissue obtained from patients could be cultured in chips to gain valuable information about tumour type, clues about possible adverse reactions, and help decide the best treatment strategy [105]. Information generated from these chips could help physicians come up with new therapeutic combinations that could reduce the use the chemotherapy. Also, a swift diagnosis would ensure the commencement of treatment before the tumour reaches the metastatic stage. The ability to create 3D tissue structures combined with the small working volumes of chips could enable the more in-depth study of the progression of cancer tumours, cell–cell interactions and expand knowledge on how tumours invade distant tissues.

4.2 Microfluidic Devices in Drug Discovery

Drug discovery is a long process, with possible drug candidates turning into successful new drugs taking over ten years. Even with recent advancements, the development process is faced with many challenges, and with the increasing costs of clinical testing, there is a need for better tools to streamline the process. There is a requirement for better tools to quickly analyse drug candidates and predict their efficacy with greater accuracy. Drug candidates showing promise often fail in preclinical and clinical stages, highlighting the inadequacy of current drug discovery techniques. Microfluidic devices can manipulate fluids using microchannels, pumps, and valves, which could significantly reduce times for high-throughput screening and make the process more economical with their small working volumes and quantities of reagents required.

4.2.1 In Target Selection

Drug action is primarily based upon the drugs' binding to specific receptors to either interrupt or induce a signal-transduction pathway. Hence it becomes essential to understand these protein–protein interactions. This process is generally begun by lysing cells and separating their protein constituents and labelling them. Microfluidic devices have been applied for this purpose using single-molecule fluorescence counting. An example of this application is the determination of the number of epitope-tagged β_2- adrenergic receptors (β_2ARs) [106]. Microfluidic devices integrated with electrophoretic devices have been applied for the separation and analysis of amino acids, thus improving basic biomedical study [107, 108].

4.2.2 In Separating Proteins and Crystallization

Receptor modelling requires the separation of the proteins and crystallization. Microfluidic devices provide several benefits, such as higher speeds, efficiency, portability, and automation. Microfluidic devices coupled with micellar electrokinetic chromatography (MEKC) and capillary zone electrophoresis have been applied for the separation of polypeptides [109]. Microfluidic devices, coupled with capillary gel electrophoresis using different buffered solutions, were applied to separate proteins. The use of on-chip valves ensured the separation of these fluids, and the entire separation process can be completed in about 20 min [110].

Crystallization of proteins has its own unique challenges because of the flexibility of macromolecules and relatively fewer hydrophilic loops that could potentially act as crystal contact points. These challenges were overcome by preparing droplets containing the protein, additives, and precipitants in immiscible fluids, resulting in a two-phase system. The protein now could be induced to crystallize inside the aqueous droplet. This two-phase system was later sheared to separate the aqueous fluids and obtain the crystals of the protein, the presence of which was confirmed using X-ray diffraction technique [111, 112]. Large scale protein crystallization has also been achieved using 480 on-chip valves and 114 parallel reactors [113]. Droplet-based microfluidics has also been applied for crystallization and successfully extract high-quality crystals of membrane proteins. The process was also highly efficient, with over 1300 crystallization trials carried within 20 min [114]. Understanding the phase behaviour of protein can increase success in crystallization processes and was applied to a microfluidic platform and was successfully able to extract 12 proteins that failed to crystallize using conventional techniques [115].

4.2.3 In Predicting Ligand Binding

The conventional method of determining ligand binding involves exposing the target protein to different drug concentrations and determining half-maximal inhibitory concentration (IC_{50}) or equilibrium dissociation constant (K_d) for the drug-target protein binary complex. Microfluidic devices can enhance the efficiency and sensitivity of this process. A high throughput microfluidic platform was designed to detect low-affinity binding. It was used to determine the binding energies of four eukaryotic Transcription factors to DNA, which also helped predict the in-vivo function of two yeast Transcription factors and determining binding specificity of the transcription factors [116]. Another application of this platform involved the use of label-free photonic crystal biosensors embedded in the bottom surface of channels of the device. It was used to measure the kinetic binding interaction of proteins. IgG molecules of low medium and high affinity were used for the experiment. It borrowed the format of a 96 well microplate allowing one common access well to 11 flow channels. Pneumatic pressure was used to drive flow through the channels. Overall, it allowed the minimization of the volume of reagent required to functionalize the biosensor surface [117]. Quantitative analysis of ligand binding involves the accurate measurement of values of binding energy, either enthalpically [118] or by NMR spectroscopy [119]. However, this information is difficult to obtain and microfluidic devices can ease this process. Magnetic nano sensors have been previously utilized for detecting molecular interactions [120], which can be applied to microfluidic devices because of their small working volumes.

4.2.4 In High-throughput Screening

High throughput screening is an automated procedure that screens a large number of compounds from various libraries capable of eliciting a biological response at the organism, cellular, pathway, or molecular level. It has been a powerful tool for pharmaceutical companies to screen new chemical entities [121]. The conventional method of HTS involves the use of multiple well plates. However, it is not easy to dispense nanolitres of fluid in these wells and the method ends up requiring large quantities of expensive reagents. Hence microfluidic devices are advantageous because of their ability to function with small volumes and allowing multiple determinations to be run in parallel.

A high-density microfluidic device was designed for this purpose. It consisted of an extensive network of channels with hundreds of individually addressable chambers and thousands of micromechanical valves and actuators capable of controlling fluid flow to these individual chambers. This system also features a fluidic multiplexor, a combinatorial array of binary valves capable of producing complex combinations of fluids. The HTS capability of this system was demonstrated using Escherichia coli expressing recombinant cytochrome c peroxidase (CCP) enzyme and dyes such as Amplex Red [122]. Microfluidic devices have also been applied for the purification of nucleic acids from bacterial and mammalian cells. The design included all steps such as isolation, cell lysis, DNA or mRNA purification, and recovery onto the same chip with no requirements of any pre- or post-treatment steps. Measurable quantities of mRNA could be extracted from a single cell, and the use of the parallel architecture of microfluidic devices gave the ability to run several samples simultaneously [123].

4.3 Microfluidic Devices in Synthetic Biology

Advances in synthetic biology have a significant impact on our abilities to generate new biomaterials and novel therapeutics. Microfluidic platforms enable a wide variety of synthetic processes to be performed either in continuous or batch format. Their low working volumes not only enhance the rates of mass and thermal transfer but also the efficiency of mixing and separation, reducing the time required to generate high-quality products. Processes such as DNA amplification, cell, and tissue culturing, immunoassays, and high throughput screening are highly successful on this platform [124]. The introduction of PCR chips has led to the development of several microfluidic-based PCR technologies offering several advantages such as faster ramping rates, better heat transfer rates, lower costs, and high integration [125]. Current de novo synthesis of DNA is expensive, with reagents and handling costs driving the price upwards, hampering the widespread application of this technology. However,

the ability to synthesize synthetic DNA constructs rapidly and accurately is highly desired by researchers. Microfluidic systems can reduce costs because of their lower requirement of reagent volumes and their ability to run several complex reactions in parallel. A multi-chamber microfluidic device was used to synthesize genes up to 1 kb in length from oligonucleotides solutions of concentrations lying in the range of 10-25 nM placed in four 500nL reactors. The total volume required for this synthesis was one to two orders less than the volumes required by conventional methods of gene synthesis. The resultant genes had a low rate of error, verified by sequencing, and the error rate was 1 per 560 bases [126].

5 Challenges

Although microfluidic platforms have demonstrated the outstanding potential of application in various fields, there remain several challenges that need to be overcome before this system can gain worldwide acceptance. Although these systems aim to mimic physiological conditions, one of the significant drawbacks this system suffers from is the liquid to cell ratio, which tends to be 10 to 100-fold higher than physiological values. This large ratio leads to dilution of the drug and its metabolites, understating the expected adverse reactions. However, reducing fluid volumes also risks cell death and could lead to volumes too low for quantitative analysis. There has been a transition from using 2D cultures to 3D cultures using scaffolds made of natural hydrogels such as gelatine, alginate, and synthetic hydrogels such as polyethylene glycol to mimic in-vivo structure better. However, these fail to mimic cell–cell and cell-extracellular matrix interactions or impede nutrient, oxygen, or waste transfer. Currently, cells derived from immortal cell lines are utilized. However, primary cell lines or stem cell-derived cells better mimic human tissues. Nonetheless, it has been challenging to use these tissues because of the lack of a common culture medium that can fulfil the requirements of different types of tissues simultaneously.

Furthermore, they cannot mimic the interstitial fluid compartment present in vivo. In-vivo, there is rapid partitioning of drugs from blood, which leads to dilution of drugs and drug metabolites. Hence it is essential to account for this fluid compartment when evaluating data from chips. Although being superior at mimicking in-vivo conditions than current in-vitro platforms, there remains ground to be covered until these chips can completely simulate in-vivo conditions. There remains the need for mathematical adjustments to predict outcomes in patients successfully. The data generated from these chips can be relied on only if they recapitulate in-vivo conditions accurately [127].

Also, the transition of these chips from laboratory to market has been slow. The causes, as mentioned earlier, can be held responsible. Although there have been advancements in chip fabrication and the development of pumpless systems, this platform's effectiveness, in the end, is dependent upon the equipment and techniques used for quantitative analysis. There is a need for highly sensitive equipment which are not widely used and is expensive to operate, driving up the running costs of this platform. Hence there is a need for a shift of attention from miniaturization to commercial applicability [128].

6 Conclusion

In this review article, we have examined a wide range of applications of organ-on-chip technology in various stages of the drug discovery process. From disease modelling, to understanding pharmacokinetics of drug molecules and being applied in early drug discovery, we can appreciate the versatility of the platform. There has been active funding from various agencies such as the National Institute of Health (NIH), Food and Drug Administration (FDA), Defence Advanced Research and Projects Agency (DARPA), and the European Union, which is a testament to the growing interest in this platform. This platform can solve several problems with current drug testing techniques and possibly replace animal testing. However, several hurdles need to be overcome before it becomes a drug testing tool for the future. Nonetheless, the growing interest of several pharmaceutical and cosmetic companies [129] to commercialize this platform will ensure the continued development and evolution of this platform.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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